

2(a)	$+\Delta U$ <u>increase</u> in internal energy $+q$ heat (energy) transferred <u>to</u> the system / heating of system $+w$ work done <u>on</u> system	B2
2(b)(i)	$W = p\Delta V$ $= 5.2 \times 10^5 \times (5.0 - 1.6) \times 10^{-4}$ (=177 J)	B1
	$\Delta U = q + w$ $= 442 - 177 = 265 \text{ J}$	A1
2(b)(ii)	no (molecular) potential energy	B1
	internal energy decreases so (total molecular) kinetic energy decreases	B1
	(mean molecular) kinetic energy decreases so temperature decreases	B1

Question	Answer	Marks
2(b)(iii)	$\Delta U + 265 - 313 = 0$ $\Delta U = 48 \text{ J}$	A1
2(b)(iv)	$pV = NkT$ or $pV = nRT$ <u>and</u> $N = nN_A$	C1
	$5.2 \times 10^5 \times 1.6 \times 10^{-4} = N \times 1.38 \times 10^{-23} \times (273 + 227)$ or $5.2 \times 10^5 \times 1.6 \times 10^{-4} = n \times 8.31 \times (273 + 227)$ and $n = N / 6.02 \times 10^{23}$ $N = 1.2 \times 10^{22}$	A1

Question	Answer	Marks
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